GoalFlow
Technical Architecture

How the system is built and what every piece is used for

Stack Flutter · Node.js + TypeScript + Express · MongoDB Atlas · Firebase Cloud Messaging · Resend

Demo login demo@goalflow.app / Demo1234

How the system is put together, what each library is doing, and why it is there. Written so another developer can change something without first reading everything.

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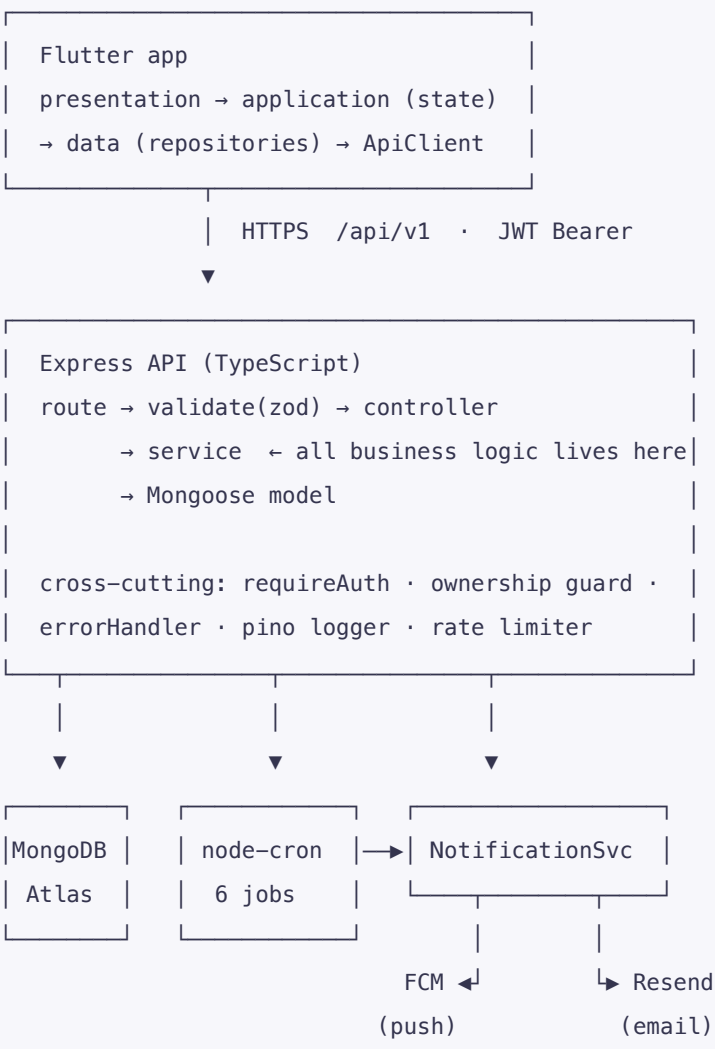
1. Shape of the system



One rule holds the whole thing together: business logic lives in services on the backend and in providers on the client. A route handler never computes anything, and a widget never calls the network.

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2. Backend libraries — what each one actually does

Library	Used for	Why this one
`express`	HTTP routing	Layering stays visible to a reviewer; NestJS would add structure this size of app does not need
`mongoose`	MongoDB ODM	Schema validation, indexes and populate in one place
`zod`	Request + env validation	Parse at the edge, infer the type — one source of truth instead of a schema plus an interface
`bcryptjs`	Password hashing	Pure JS, no native build step, so `npm install` never fails on a fresh machine
`jsonwebtoken`	Access tokens	Stateless auth; refresh tokens are opaque and stored hashed
`luxon`	Timezone maths	Every schedule decision is per-user-local; `Date` cannot do this correctly
`node-cron`	Scheduled jobs	Six jobs, in-process. Redis/BullMQ would be right at multi-instance scale, not here
`firebase-admin`	Push (FCM)	Backend-driven sends to Android and iOS from one API
`resend`	Transactional email	Verification, reset, weekly digest
`pino` + `pino-http`	Structured logging	Request IDs, and redaction so passwords never reach a log
`helmet`, `cors`, `express-rate-limit`	HTTP hardening	Standard headers, and brute-force limits on credential routes
`multer`	Avatar upload	Multipart parsing; images are stored in Mongo, not on Render's ephemeral disk

Request lifecycle

```
POST /api/v1/goals
 → helmet, cors, json body parse
 → pino-http           attaches a request id, logs start/finish
 → requireAuth         verifies the JWT, loads the user onto req.user
 → validate(schema)    zod parses body/query/params, or throws field errors
 → controller          unwraps the request, calls the service, shapes the response
 → service             the actual work: create the goal, recompute status
 → Mongoose model      writes to MongoDB
 → ok(res, data)       { success: true, message, data }
 → errorHandler        anything thrown lands here and becomes one error shape
```

Every success looks like `{ success, message, data }`. Every failure looks like `{ success: false, error: { code, message, details? } }`. The client parses exactly one shape, and `details` carries per-field messages so a form can highlight the offending input.

Environment

`src/config/env.ts` runs every variable through zod at boot. A missing or malformed value **exits the process with the key name printed**, rather than failing later on a random request. This is what surfaced the missing JWT secrets during the first Render deploy.

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3. Data model

Seven collections. Everything is scoped by `user`.

```
User ─┬─< Goal ─┬─< Milestone ─< Action ─< ActionOccurrence
      │         └─< Action (goal-level, milestone optional)
      ├─< WeeklyReflection
      ├─< Notification
      ├── preferences            (embedded)
      ├── notificationPreference (embedded)
      ├── refreshTokens[]        (hashed, select:false)
      └── deviceTokens[]         (FCM)
```

The plan/log split

This is the decision everything else depends on.

	Role	Example
`Action`	the **rule**	"Practise Spanish, 25 min, Mon–Fri, 07:30"
`ActionOccurrence`	one **dated instance**	"Wed 13 Aug, completed at 07:52"

A recurring action has no single status — "did I miss Monday?" is unanswerable if the action is one row. Splitting them makes it a lookup:

Feature	Query
Today's actions	`{ user, scheduledDate: today }`
Missed detection	cron sets `upcoming → missed` once the user's day ends
Calendar	occurrences already carry a date
Weekly progress	count planned vs completed in the week
Streak	walk occurrence days backwards
Reminders	`scheduledAt` **is** the queue

Occurrences are never deleted, only status-changed, which is how history survives.

Indexes

Collection	Index	Purpose
`actionoccurrences`	`{action, scheduledDate}` **unique**	Makes materialisation idempotent — the cron can run twice without duplicating a row
`actionoccurrences`	`{user, scheduledDate, status}`	Today's feed, calendar range
`actionoccurrences`	`{status, scheduledAt}`	The 15-minute reminder sweep
`actionoccurrences`	`{goal, scheduledDate}`	Goal detail and progress
`goals`	`{user, status, targetDate}`	Goals list
`users`	`{email}` unique	Login
`weeklyreflections`	`{user, weekStart}` unique	One reflection per week

4. The scheduler

`modules/occurrences/materialiser.service.ts` turns rules into dated rows.

```
for each active goal
  for each active action
    work out which local days it falls on (daily | specific_days | weekly_count | once)
    resolve the time:  action.preferredTime → goal.routine.startTime → user preference
    upsert one occurrence per day, 7 days ahead
```

Three details that matter:

- **Timezones.** Days are computed in `user.timezone` via luxon, then stored as UTC. `scheduledDate` is the UTC instant of *local* midnight, so a day-bucket query is a plain range scan and never depends on the device clock.
- **Idempotent.** `$setOnInsert` plus the unique index means re-running changes nothing.
- **Editing a routine rewrites only the future.** Upcoming occurrences from today onward are deleted and regenerated; completed and missed history is never touched.

The six cron jobs

Job	Cadence	Does
`materialise`	hourly at :10	Generate the next 7 days for every user
`mark-missed`	hourly at :20	`upcoming → missed` once a user's local day has ended
`action-reminders`	every 15 min	Push for occurrences due within the user's lead time
`daily-summary`	every 15 min	Fires in the bucket containing the user's chosen time
`weekly-digest`	every 30 min	Push + Resend email on the user's chosen weekday/time
`recompute-goals`	01:00 daily	Refresh cached `progressPercent` and `computedStatus`

Jobs run **hourly, not daily**, and compare against each user's *local* clock — that is how one server serves every timezone without per-user cron entries. Each job is wrapped in a re-entrancy guard so a slow run never overlaps itself on a small instance.

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5. Progress status algorithm

`modules/goals/goal.status.service.ts`. Deliberately not a percentage.

```
timeRatio = elapsed / total goal window
workRatio = completed occurrences / all planned occurrences to the target date
adherence = last 14 days: completed / (completed + missed)     [skips excluded]
delta     = workRatio − timeRatio
```

First match wins:

Condition	Status
`workRatio >= 1` or marked complete	**Completed**
goal younger than 7 days	**On track** — grace period
`delta >= +0.10`	**Ahead**
`delta >= -0.05` and `adherence >= 0.60`	**On track**
`delta >= -0.20` or `adherence >= 0.40`	**Needs attention**
otherwise	**Behind**

Notes on the choices:

- **Skips are excluded from adherence.** A skip is a decision; a miss is a failure. Counting them the same punishes honesty.
- **The denominator is projected.** Occurrences only exist 7 days ahead, so the remaining calendar is projected at the goal's own weekly cadence — otherwise every goal would look 100% complete.
- **Every status carries a sentence.** `statusReason` is generated with the status and rendered verbatim. A coloured chip alone explains nothing.

Consistency (`modules/progress/progress.service.ts`) counts a day only if something was planned for it. Days with nothing planned are *neutral* — they neither break nor extend a streak. A rest day you scheduled yourself is not a failure.

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6. Notifications

The pipeline

```
cron job / login / milestone
        │
        ▼
 NotificationService.dispatch()
        │
        ├── is this type enabled for this user?
        ├── is it inside their quiet hours?   (overnight ranges handled)
        │
        ├──▶ sendPush()   FCM multicast → prunes tokens FCM rejects
        ├──▶ sendEmail()  Resend, or logs the payload in dry-run
        └──▶ Notification document → the in-app feed and delivery audit
```

Nothing sends outside this function. Controllers and jobs never touch FCM or Resend directly, which is why "respects user preferences" is a property of the system rather than something each caller has to

remember. Transactional mail (verification, password reset) passes `bypassPreferences` — you cannot opt out of your own reset link.

Firebase — deliberately limited

Firebase is used for **push only**, plus verifying a Google sign-in ID token. Sessions are always issued by this backend. The service account is *project*-level, so one Firebase project can serve several apps; only each mobile app needs registering separately, because `google-services.json` is tied to a package name.

Both sides degrade safely: without credentials the backend logs the payload instead of sending, and the app runs with push disabled rather than crashing at launch.

Login greeting

`modules/notifications/greeting.service.ts` — the same canned "Welcome back" every time makes an app feel like it isn't paying attention, so the message is chosen from the gap since the previous sign-in:

Gap	Kind	Message
never signed in	`first_time`	*Welcome to GoalFlow, Priya* — set your first goal
same day	`same_day`	*Hey again, Sam* — 4 things still open today
1–3 days	`returning`	*Welcome back, Sam* — 3 actions planned today, starting with Learn Spanish
4–14 days	`been_a_while`	*Good to see you, Sam* — it has been 9 days, pick one and you are moving again
15+ days	`long_absence`	*Welcome back, Sam* — 40 days away, no catching up needed, just start from today

Three implementation details:

- **Token refresh does not greet.** `issueSession` takes `{greet: false}` from `refresh()`, otherwise a notification would fire every 15 minutes.
- **First-ever login has no device token yet.** The greeting is stored and flagged `pendingGreetingAt`; registering a device flushes it (if still fresh) so the one greeting that matters most is not lost.
- **Failures are swallowed.** The whole thing is wrapped — a greeting must never be able to break signing in.

Platform setup

Platform	Needed
Android	`POST_NOTIFICATIONS` (13+), default FCM channel `goalflow_default` declared in the manifest, `google-services.json` in `android/app/`
iOS	`remote-notification` background mode, `Runner.entitlements` with `aps-environment`, `GoogleService-Info.plist` in `ios/Runner/`, APNs key uploaded to Firebase

The client handles all three delivery states: foreground (rendered via `flutter_local_notifications`, since iOS shows nothing by default), background (system tray), and terminated (`getInitialMessage` on launch). Tapping any of them reads `data.route` and navigates.

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7. Authentication

Concern	Approach
Password storage	bcrypt, cost 12, `select: false` so it is never returned by accident
Access token	JWT, 15 minutes
Refresh token	48 random bytes, stored **SHA-256 hashed** — a database dump cannot be replayed as a session
Rotation	The presented refresh token is consumed before a new one is issued
Session limit	Newest 5 kept per account
Password change/reset	Clears every refresh token — all devices sign out
Account enumeration	Login, forgot-password and resend-code give identical answers for known and unknown emails
Brute force	Rate limiter on every credential route
Authorization	Ownership is checked per resource, not assumed from a query filter

Client side: tokens live in the platform keychain via `flutter_secure_storage`. A 401 triggers **one** transparent refresh-and-retry inside `ApiClient`; if that fails the session is cleared and the router drops to login. No screen implements this.

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8. Flutter app

Library	Used for	Why
`flutter_riverpod`	State + dependency injection	Compile-safe, testable without a widget tree, keeps logic out of widgets
`go_router`	Routing	Declarative, one auth redirect guard, deep links for free
`dio`	HTTP	Interceptors give one place for the auth header, refresh-retry and error mapping
`flutter_secure_storage`	Tokens	Keychain / Keystore, not SharedPreferences
`shared_preferences`	Theme choice	Non-sensitive device preference
`firebase_messaging` + `flutter_local_notifications`	Push	Remote delivery, and foreground rendering iOS otherwise suppresses
`fl_chart`	Progress charts	Only on the Progress screen
`table_calendar`	Schedule grid	Month view with per-day markers
`intl`	Dates and durations	Locale-aware formatting

Layers

```
presentation/   widgets and screens. No network calls, no calculations.
application/    Riverpod providers and notifiers. All client-side logic.
data/           models (hand-written fromJson) + repositories.
core/           theme, router, ApiClient, secure storage, push service.
```

No codegen. Models use hand-written `fromJson`, so a fresh clone runs with `flutter pub get` and nothing else — no `build_runner` step to forget.

How a tap flows

```
user taps the checkbox on Today
 → widget calls occurrenceActionsProvider.complete(id)
 → OccurrenceActions calls AppRepository.complete()
 → ApiClient POSTs /occurrences/:id/complete, unwraps the envelope
 → on success, invalidates dashboard, today, goals, progress, calendar, reflection
 → every affected screen refetches
```

One invalidation list is why ticking an action on Today is immediately visible on Progress and in the Calendar, with no manual cache juggling.

API URL resolution

```
--dart-define=API_BASE_URL=...     explicit, always wins
--dart-define=USE_LOCAL_API=true   localhost (10.0.2.2 on an Android emulator)
otherwise                          the deployed backend
```

Production is the **default** on purpose: an APK handed to someone else must work with no flags. Timeouts are 75 seconds because a free-tier host cold start was measured at 17 s and can reach a minute — a shorter timeout reads as a broken app.

Theming

`AppTheme.light()` / `AppTheme.dark()` are built from one function, so a colour is never defined in only one mode. Spacing, radii and type come from the `Gap` scale rather than magic numbers. The chosen mode is persisted and applied instantly — state changes first, storage is written after, so the toggle never feels laggy.

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9. Testing and verification

```
cd goalflow-api && npm run typecheck   # tsc --noEmit, clean
cd goalflow_app && flutter analyze     # no issues
cd goalflow_app && flutter test        # 12 tests
```

Verified against the deployed stack: login, dashboard, complete and undo an action, progress summary, weekly reflection, calendar range, auth guard, all five greeting variants, and FCM credential authentication.

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10. Known limits

Limit	Detail
Render free tier sleeps	Cron does not run while asleep. Fix: paid instance, or an external `/health` ping. The app also regenerates missing days on open, and Settings has *Rebuild my schedule*
Push needs per-app registration	`google-services.json` is bound to a package name; the backend service account is project-wide and already reusable
iOS push needs an Apple developer account	An APNs key must be uploaded to Firebase
Cron is in-process	Correct for one instance. Multiple instances would need a distributed lock or a queue
Occurrences grow over time	7 days ahead only, indexed. Old rows would eventually want archiving